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Basic science

High levels of EPSTI1 enhance IFN-β-mediated HLA-A expression and chemokine secretion in myoblasts in dermatomyositis

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Abstract

Objectives: Epithelial-Stromal Interaction 1 (EPSTI1), an IFN-related gene that has emerged as a gene of notable interest, plays a multifaceted role in cellular function and disease processes. However, the precise role of EPSTI1 in the context of DM remains elusive and requires further exploration.

Methods: To investigate EPSTI1 expression in DM, we analysed two transcriptome datasets, peripheral blood mononuclear cells (PBMCs) and muscle tissues from our DM cohort. We further validated the findings using RT-qPCR and immunohistochemical staining within our idiopathic inflammatory myopathy (IIM) cohort. Through small interfering RNA–mediated knockdown, we delved into the function and underlying mechanisms of EPSTI1 in DM patients.

Results: EPSTI1 is significantly upregulated in both PBMCs and muscle tissues of DM patients. A notable positive correlation between EPSTI1 expression and IFN-stimulating genes accompanies this upregulation. Furthermore, EPSTI1 levels were markedly elevated in IFN-β-stimulated myoblasts. Functional studies have shown that downregulation of EPSTI1 inhibits multiple immune and inflammatory pathways, including antigen processing and presentation, chemokine-mediated signalling, IFN signalling, IL-1-mediated signalling, nuclear factor (NF)-κB-inducing kinase (NIK)/NF-κB signalling. EPSTI1 was involved in the expression of HLA-A and the secretion of chemokines in myoblasts mediated by IFN-β.

Conclusion: EPSTI1 might play an essential role in promoting muscle inflammation in DM by regulating the expression of HLA-A and the secretion of chemokines in myoblasts. Targeting EPSTI1 may represent a novel therapeutic approach for reducing muscle inflammation and damage in dermatomyositis (DM) patients.

Keywords: dermatomyositis, EPSTI1, interferon, myoblasts.

Rheumatology key messages

- Epithelial-Stromal Interaction 1 (EPSTI1) is specifically highly upregulated in both peripheral blood mononuclear cells and muscle tissues of DM patients.
- MHC class I (MHC-I), MHC-II and chemokines genes were elevated in IFN-β-stimulated myoblasts, which was inhibited by knocking down ESPTI1.
- EPSTI1 was involved in the expression of HLA-A and the secretion of chemokines.

Introduction

Idiopathic inflammatory myopathy (IIM) encompasses a spectrum of autoimmune disorders with significant heterogeneity, primarily comprising DM, anti-synthetase syndrome (ASyS), immune-mediated necrotizing myopathy (IMNM), PM, sporadic IBM (sIBM) and overlap myositis [1, 2]. As the most prevalent subtype of IIM, DM is characterized by proximal myalgia, myasthenia and a distinctive rash. Additionally,

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DM patients may exhibit extra-muscular manifestations such as pulmonary interstitial disease and arthritis [3, 4]. The muscle pathology of DM is characterized by several distinct features, including peribundle atrophy, marginal vacuoles, positive staining for MxA (a marker of viral infection and IFN response) and inflammatory cell infiltration [5]. Notably, the expression patterns of primary MHC class I (MHC-I) and class II (MHC-II) molecules in DM muscles are unique and distinguish them from other muscular dystrophy lesions [6, 7]. Despite these well-documented pathological features, the underlying mechanisms responsible for the development of DM remain incompletely understood.

The IFN signalling pathway is generally activated in the muscles, peripheral blood and skin of DM patients [8–10]. This activation leads to the upregulation of IFN-stimulating genes (ISGs), including ISG15, RIG-I, MxA and the IFN-related chemokine CXCL10, in both muscle tissue and peripheral blood transcripts of DM patients [11–13]. Additionally, the IFN-regulated genes (IRGs) score in peripheral blood correlate with disease activity scores [14]. Recently, a multicentre trial demonstrated that dazukibart, which is a humanized IgG1 neutralizing monoclonal antibody directed against IFN-β, resulted in a pronounced reduction in the Cutaneous Dermatomyositis Disease Area and Severity Index Activity (CDASIA) score, and improvements in the total improvement score (TIS), manual muscle testing score (MMT8), patient global assessment (PtGA) and creatine kinase [15], suggesting the involvement of the IFN signalling pathway in the pathogenesis of DM.

In this study, we screened up-regulated differentially expressed genes (DEGs) from two independent skeletal muscle transcriptome datasets {our DM muscle transcriptional [48 DM, 10 normal controls (NC)] cohort and GSE143323 (39 DM and 20 NC)}. Subsequent fold-change (FC) ranking of these DEGs identified the top 100 candidate genes from each dataset. Venn analysis revealed 21 overlapping upregulated genes (Supplementary Fig. S1A, available at *Rheumatology* online). Given the systemic nature of DM pathogenesis, we integrated these 21 genes and upregulated DEGs derived from two PBMC transcriptomic data sets [our DM PBMC transcriptional (16 DM, 12 NC) cohorts and GSE125977 (7 DM and 5 NC)], yielding 12 DEGs, all of which are IFN-stimulated or related genes. Some of these genes have been reported in myositis, such as ISG15, HERC5, MX1, RSAD2 and IFI family genes [16–19]. The Epithelial-Stromal Interaction 1 protein (EPSTI1), categorized as an ISG, plays a multifaceted role in cellular function and disease processes. It is primarily localized in the cytoplasm and is partially expressed in the cell membrane and nucleus. In the initial discovery, EPSTI1 was significantly induced by Epithelial-Stromal Interaction 1 [20, 21]. In JDM patients, a set of IRGs composed of EPSTI1 and others have been reported to be associated with disease activity [22]. However, the exact role of EPSTI1 in DM and its relationship with IFN-related genes remains an area of ongoing research and investigation.

In this research, we first assessed the expression levels of EPSTI1 in peripheral blood mononuclear cells (PBMCs) and muscle tissues of patients with DM. Furthermore, we conducted experiments by stimulating myocytes with IFN to explore the functional significance and underlying mechanisms of EPSTI1 in DM.

Methods

Patient enrolment

In this study, we used two transcriptional cohorts and one validation cohort to reveal the expression of EPSTI1 in DM patients. The inclusion criteria for DM patients and NC in the two transcriptional cohorts were based on our previous publication [23]. Blood samples were collected from 16 DM and 12 age- and sex-matched NC in the first transcriptional cohort (Supplementary Table S1, available at *Rheumatology* online). Muscle samples were collected from 48 DM and 10 NC in the second transcriptional cohort (Supplementary Table S2, available at *Rheumatology* online). The RT-PCR validation cohort includes 40 DM, 5 IMNM, 5 ASyS and 24 controls (Supplementary Table S3, available at *Rheumatology* online). The patients were recruited from the Department of Rheumatology and Immunology at Xiangya Hospital, Central South University, between 2018 and 2022. They were respectively diagnosed according to the 2019 European Neuromuscular Centre (ENMC) criteria, 2018 ENMC criteria and Connor criteria [24–26]. All patients were newly diagnosed with IIM and had not received glucocorticoid or immunosuppressant treatment. MSAs were tested using Euroline Myositis Profile immunoline-blot (Euroimmun), and more than one strongly positive MSA result was excluded. Other exclusion criteria included infection, tumour, severe cardiovascular or metabolic disease, overlap syndrome and sIBM. Control muscle biopsies were obtained from individuals in a stable phase with osteonecrosis of the humeral head and OA undergoing shoulder replacement surgery.

Transcriptional profiling analysis of PBMCs from DM and NC

PBMCs were separated from heparinized blood (16 DM, 12 NC) by density gradient centrifugation using Ficoll-Paque Plus. Total RNA was extracted using a Trizol reagent, and its concentration was measured using a Nanodrop ND-1000 Spectrophotometer. RNA quality was assessed using an Agilent Bioanalyzer, with only samples having an RNA Integrity number >7 used for microarray analysis. Illumina Human HT-12 v4.0 Expression Beadchips (Illumina Inc., San Diego, CA, USA) was used for mRNA transcription profiling, containing 47 323 transcripts. RNA samples were amplified using the Illumina Total Prep RNA Amplification Kit (Thermal Fisher Scientific) with biotin-UTP (Enzo Life Sciences, Inc., Farmingdale, NY, USA) labelling. The process involved generating single-stranded cDNA with T7-oligo(dT) primers, followed by second-strand synthesis, purification and biotin-labelled cRNA synthesis using T7 RNA polymerase. The cRNA was quantified using an ND-1000 Spectrophotometer (Thermo Fisher Scientific), and 1.5 μg of cRNA was hybridized on each array with streptavidin-Cy3 (Amersham Biosciences Corp., Piscataway, NJ, USA) detection. Arrays were scanned using an Illumina HiScan scanner and analysed with GenomeStudio (Illumina Inc., San Diego, CA, USA). Data were log2-transformed and normalized using the lumi package in R. Only 'perfect' and 'good' annotated probes were included. Quality control revealed no batch effects. Differential gene expression between disease and

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normal groups was analysed using t-tests with Benjamini–Hochberg multiple comparisons correction, with statistical significance defined as q-value <0.05 and FC ≥1.5. Visualization was performed using R packages ggplot2, ggpubr and volcano plots.

Bulk RNA-seq of muscle tissues from patients and cultured myoblasts

Total RNA was extracted from muscle tissues (48 DM, 10 NC) and myoblasts (including IFN-β stimulation and EPSTI1 knockdown) using Trizol reagent according to the manufacturer's protocol. RNA sequencing (RNA-seq) was performed by Novogene (Beijing, China) on an Illumina NovaSeq platform using a paired-end 150 bp sequencing strategy. The quality of the sequencing reads was assessed using FastQC (v0.11.8), and clean reads were obtained using Trimmomatic (v0.38). The clean reads were aligned to the human reference genome (hg38) using hisat2 (v2.1.0), and the alignment was managed with Samtools (v1.9). Gene annotation was performed using subread (v1.6.3) with the Gencode v35 annotation file. Count data were normalized using the TMM method, and DEGs were identified using edgeR (v3.32.0) with criteria of FC >1.2 and *P*-value <0.05. Gene functional analysis was conducted using clusterProfiler. The Visualization was performed using R packages ggplot2, ggpubr, volcano plots, clusterprofiles and pheatmap.

Gene expression omnibus datasets analysis

We selected the datasets with 'dermatomyositis' from the Gene Expression Omnibus (GEO) database (https://www.ncbi.nlm.nih.gov/geo/). GSE125977 (whole-blood RNA sequencing in 7 DM and 5 NC) and GSE143323 (muscle tissue RNA sequencing in 39 DM and 20 NC) have been selected. We performed quality control checks and normalization on the downloaded data. The R packages limma and DESeq2 were employed to identify significant DEGs (log2FC > 1.5, $P < 0.05$). Visualization was performed using R packages ggpubr, ggplot2 and volcano plots.

Quantitative real time-PCR

Total RNA was extracted utilizing the TRIzol reagent as the manufacturer's instructions. The concentration and quality of RNA were assessed using a QS3000 spectrophotometer. For qRT-PCR, gene-specific primers were employed with SYBR Green (SYBR Premix Ex Taq RT-PCR kit, Takara) on a 7500 Real-Time PCR System (Applied Biosystems, Waltham, MA, USA). The primer sequences can be found in Supplementary Table S4, available at *Rheumatology* online. ACTIN was employed as the internal reference, and the $2^{-\Delta CT}$ method was used to quantify the relative expression of mRNA.

Immunohistochemical analysis

The immunohistochemistry was performed similarly to the previous paper published by our group [27]. Frozen muscle sections were restored to room temperature and fixed with 4% paraformaldehyde for 15 min. Endogenous peroxidase activity was inhibited with 3% H_2O_2 for 20 min, and nonspecific binding was blocked with 5% BSA for 30 min. The slides were then incubated overnight at 4°C with anti-EPSTI1 antibody (11627-1-AP, Proteintech) at a dilution ratio of 1:200. Subsequently, the slides were treated with enzyme-labelled goat anti-rabbit antibodies (Servicebio, China) at room temperature for 1 h. Following this, they were visualized using a diaminobenzidine kit as a chromogenic agent. All slides were observed using an Olympus microscope and images were captured at 20× magnification with a digital camera.

Western blot analysis

Total protein was extracted from the myoblast using RIPA lysate buffer supplemented with a protease/phosphatase inhibitor cocktail (Beyotime, Shanghai, China). BCA protein detection kits (Cat#. BCA02, DingGuo, China) were used to determine the protein concentrations. The exact amounts of protein were separated using 10% SDS–PAGE and transferred to polyvinylidene fluoride membranes. The membranes were blocked with 0.1% Tween's Tris-buffer saline containing 5% skim milk for 2 h at room temperature followed by incubation overnight with specific primary antibodies against EPSTI1 (1:500, 11627-1-AP, Proteintech), B2M (1:1000, sc-13565, Santa), HLA-A (1:1000, ab52922, Abcam) and VINCULIN (1:5000, V9131, Sigma) at 4°C. Then, the anti-rabbit IgG or anti-mouse IgG enzyme-conjugated secondary antibody (Proteintech, Wuhan, China) was incubated at room temperature for 1 h. Enhanced chemiluminescence (Bio-RAD, Hercules, CA, USA) was used to detect western blot results.

Cell culture, stimulation and transfection

Human primary skeletal muscle myoblast (ScienCell, USA) were cultured in skeletal muscle cell culture medium (SKMCM) (ScienCell, USA) supplemented with 5% FBS, 1% Skeletal muscle cell growth factor and 1% penicillin/streptomycin at 37°C with 5% CO_2. For all experiments, myoblast within passages 4–10 were utilized. Myoblast cells were transfected with 50 nM si-EPSTI1 (RiboBio, China) according to the manufacturer's protocol using Neon Electroporation System (Thermo Fisher, USA). Myoblasts were transfected with negative control small interfering RNA (siRNA) (si-NC) (RiboBio, China) served as the control group. After electrotransformation, myoblasts were cultured in SKMCM supplemented with 5% FBS, 1% Skeletal muscle cell growth factor for 24 h, and then treated with IFN-β (10 ng/ml) (PeproTech, USA) for 24 h.

Enzyme-linked immunosorbent assay

Skeletal muscle cells were cultured and treated as described above; after being treated with IFN-β for 24 h, the level of chemokines CXCL10 and CXCL11 in cells culture supernatants were determined using ELISA kit (Cloud-Clone, China) following standard instructions, and read at 450 nm. The amount of secreted cytokines was expressed as pg/ml.

Statistical analysis

R(4.2.1) software and Graphpad prism9 software were used for statistical analysis and mapping. Data conforming to the normal distribution were presented as mean ± S.D., while data not conforming to the normal distribution were presented as median (quartile). Independent sample *t*-test or Mann–Whitney *U* test was used for between-group comparisons, and one-way analysis of variance or Kruskal–Wallis H test was used for multi-group data. Spearman correlation analysis was employed to examine correlations. Statistical significance was set at $P < 0.05$.

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Ethics

This study's human experimentation protocol was approved by the ethics committee of Xiangya Hospital (No. 201212074) and conducted anonymously. All patients and controls agreed to participate in the study and provided written informed consent. All procedures involving human participants were performed following the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Results

EPSTI1 expression is upregulated in PBMCs of DM patients

We initially assessed the expression of EPSTI1 in the RNA transcriptome data of PBMCs from our DM cohort (including 16 DM patients and 12 NC). Our findings revealed a significant upregulation of EPSTI1 and ISGs (IFI family, OAS family and ISG15) (Fig. 1A). Consistently, similar findings were replicated in a publicly available DM cohort (GSE129577), consisting of 7 DM patients and 5 NC (Supplementary Fig. S2A, available at *Rheumatology* online). Furthermore, a significant positive correlation was observed between EPSTI1 and the ISGs (Fig. 1B). Compared with controls, EPSTI1 was detectable at a high level in the PBMCs of DM, ASyS and IMNM in our validation cohort (Fig. 1C). Likewise, the level of *EPSTI1* mRNA expression in anti-MDA5, anti-TIF1-γ, anti-NXP2 and anti-MI-2 patients remained significantly elevated compared with NCs (Fig. 1D). Moreover, correlation analysis unveiled a noteworthy negative correlation between EPSTI1 expression levels and MMT8 scores in all IIM patients, DM patients (Supplementary Fig. S3A, available at *Rheumatology* online) and anti-MDA5 patients (Fig. 1E).

EPSTI1 expression is upregulated in the muscles of DM patients

Consistent with the findings from the PBMC transcriptome data, the expression of EPSTI1 and ISGs, including IFI family, OAS family, STAT1 and ISG15, were upregulated in the muscle tissue of DM patients in our cohort (including 48 DM patients and 10 NC) (Fig. 2A) and in a public cohort (GSE143323, including 39 DM patients and 20 NC) (Supplementary Fig. S2B, available at *Rheumatology* online). Furthermore, correlation analysis also revealed a significant positive association between the level of EPSTI1 expression and ISGs, suggesting a potential role for EPSTI1 in IFN-mediated immune responses in DM (Fig. 2B). We subsequently conducted a validation test on our cohort. Intriguingly, the expression of EPSTI1 in the muscle tissue clearly discriminated DM from ASyS and IMNM (Fig. 2C). Likewise, the level of *EPSTI1* mRNA expression in anti-MDA5, anti-TIF1-γ and anti-NXP2 DM patients remained significantly elevated compared with NC. There was no significant difference between anti-Mi2 patients with NC (Fig. 2D). Moreover, the mRNA expression level of *EPSTI1* in muscle tissue exhibited a positive association with CK-MB levels in patients (Fig. 2E), but showed no significant association with MMT8 (data not shown). The expression of EPSTI1 in PBMCs and muscles is positively correlated in all IIM, DM and anti-MDA5 patients (Supplementary Fig. S3B, available at *Rheumatology* online). In DM, EPSTI1 expression was predominantly observed in perifascicular atrophy, blood vessels and regions infiltrated by inflammatory cells, indicating its involvement in muscle pathology (Fig. 2F).

Knocking down of EPSTI1 inhibited IFN-β-induced antigen processing presentation and chemokine-mediated signalling pathways in myoblasts.

To mimic the impact of cytokines on muscle cells *in vitro*, we treated human skeletal muscle myoblasts with a range of increasing concentrations of IFN-β and TNF-α (0, 1, 10, 30, 50, 80 and 100 ng/ml) for 24 h. Both cytokines evoked significantly increased expression of EPSTI1, with IFN-β exhibiting a more potent stimulatory effect (Supplementary Fig. S4A and B, available at *Rheumatology* online). A plateau in EPSTI1 expression was reached starting from IFN-β (10 ng/ml), which was utilized in the following experiments.

To further investigate the function of EPSTI1 in myoblasts, we silenced EPSTI1 expression in human myoblasts via transfecting siRNA and then stimulated them with IFN-β for 24 h, followed by RNA sequencing analysis. The sequencing results indicated that the stimulation of IFN-β led to an upregulation of MHC-I molecules (HLA-A, HLA-B, HLA-C, B2M) and MHC-II molecules (HLA-DOA, HLA-DOB, HLA-DMA, HLA-DMB, HLA-DRA, HLA-DRB), as well as chemokines such as *CXCL10* and *CXCL11* genes, which was inhibited by knocking down ESPTI1 (Fig. 3A). Gene set enrichment analysis revealed significant enrichment of the antigen processing and presentation, chemokine-mediated signalling, IFN signalling, IL-1-mediated signalling, nuclear factor (NF)-κB-inducing kinase (NIK)/NF-κB signalling and cell death regulation pathway following IFN-β stimulation (Fig. 3B). Downregulation of these pathways upon EPSTI1 knockdown was also significant, as shown in Fig. 3B. These findings suggest that EPSTI1 may play a crucial role in modulating the immune response in myoblast.

EPSTI1 in myoblasts induced by IFN-β promotes the expression of HLA-A and the secretion of chemokines in myoblasts

To further validate the RNA sequencing data, we conducted RT-PCR. As depicted in Fig. 4A, knocking down EPSTI1 markedly reversed IFN-β induced upregulation of MHC-I molecules and MHC-II molecules, as well as chemokines CXCL10 and CXCL11. Furthermore, EPSTI1 knockdown decreased the IFN-β-induced HLA-A protein level (Fig. 4B). The MHC-I molecule B2M component showed a similar trend, albeit less pronounced than the difference observed in HLA-A expression (Fig. 4A and B). In addition, a notable decrease of chemokines CXCL10 and CXCL11 in the cell supernatant was observed following EPSTI1 knockdown (Fig. 4C). Taken together, these findings suggest that EPSTI1 may be implicated in the process of muscle cell antigen presentation and the expression and secretion of chemokines such as CXCL10 and CXCL11, thereby facilitating the recruitment of inflammatory cells, such as monocytes, for infiltration.

Discussion

In this study, we performed transcriptome analysis of PBMCs and muscle tissues form DM patients and found that several ISGs were up-regulated, including IFIT1, IFIT2, IFIT3, IFI6, OAS1, OAS3, OASL and ISG15, as well as EPSTI1. We further validated the upregulation of EPSTI1 in our DM cohorts.

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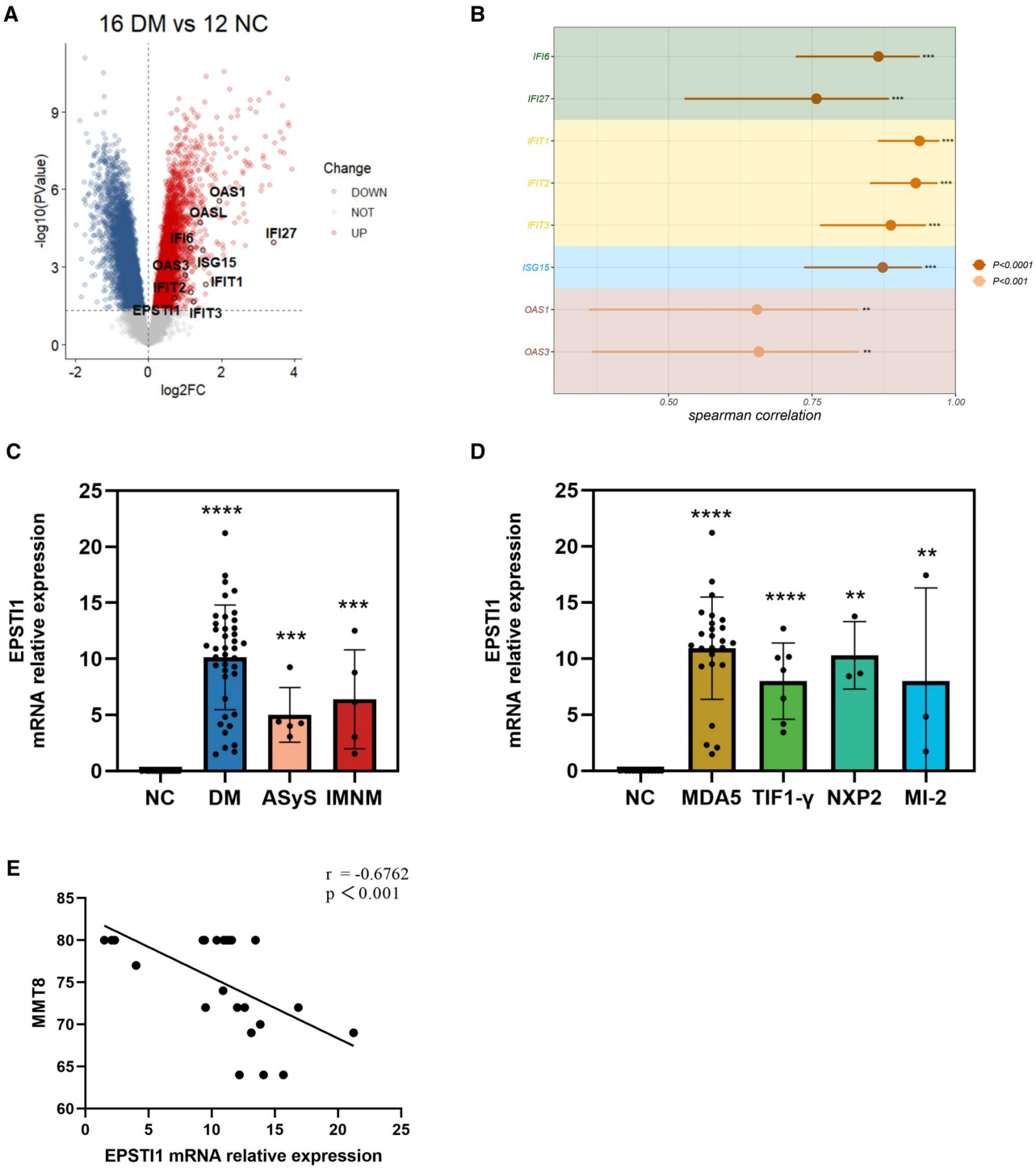

Figure 1. The expression of EPSTI1 in PBMCs of DM patients and its correlation with IFN-regulated genes and clinical indices. (**A**) Volcano plot of DEGs including EPSTI1 and IFN-stimulated genes (IFI family, OAS family and ISG15) in the RNA sequencing dataset of PBMCs from DM patients. (**B**) The correlation analysis of EPSTI1 with IFN-stimulated genes (IFI6, IFI27, IFIT1, IFIT2, IFIT3, ISG15, OAS1, OAS3) expression levels in the RNA sequencing dataset of PBMCs from patients with DM. (**C**) The relative mRNA expression of *EPSTI1* in PBMCs of patients with normal control and different clinical types of IIMs. (**D**) The relative mRNA expression of EPSTI1 in PBMCs from patients with different myositis-specific antibody groups. (**E**) The correlation between the relative expression level of *EPSTI1* mRNA in PBMCs of anti-MDA5 DM patients and MMT8. EPSTI1: Epithelial-Stromal Interaction 1; PBMCs: peripheral blood mononuclear cells; DEGs: differentially expressed genes; NC: normal controls; IIMs: idiopathic inflammatory myopathies; ASyS: anti-synthetase syndrome; IMNM: immune-mediated necrotizing myopathy; MMT8: a standardized manual muscle testing score for 8 muscle groups; MDA5: anti-melanoma differentiation gene 5; TIF1-γ: anti-transcriptional mediator 1-γ (TIF1-γ); NXP2: anti-nuclear matrix protein 2; Mi-2: anti-nucleosome remodelling deacetylase complex. In the volcano plot, upregulated genes are represented by red, while downregulated genes are represented by blue. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$

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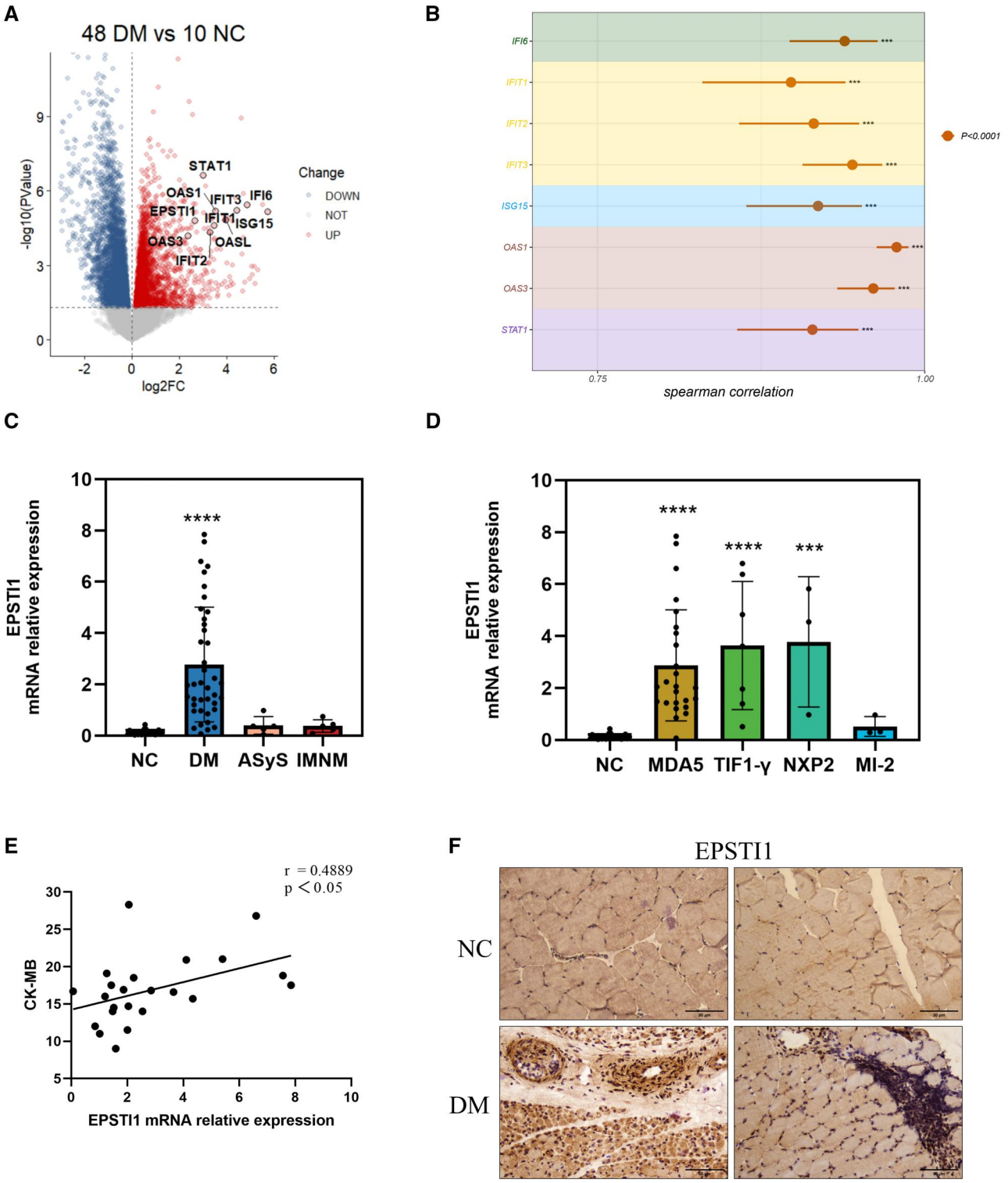

Figure 2. The expression of EPSTI1 in muscles of DM patients and its correlation with IFN-regulated genes and clinical indices. (**A**) Volcano plot of DEGs including EPSTI1 and IFN-stimulated genes (IFI family, OAS family and ISG15) in the muscle tissue RNA sequencing dataset from DM patients. (**B**) The correlation analysis of EPSTI1 with IFN-stimulated genes (IFI6, IFI27, IFIT1, IFIT2, IFIT3, ISG15, OAS1, OAS3) expression levels in the RNA sequencing dataset of muscle tissue from patients with DM. (**C**) The relative expression of *EPSTI1* mRNA in muscle tissue of patients with normal control and IIMs and different clinical types of IIMs. (**D**) The relative mRNA expression of *EPSTI1* in muscle tissue from patients with different myositis-specific antibody groups. (**E**) The correlation between the relative mRNA expression level of EPSTI1 in muscle tissue of anti-MDA5 DM patients and CK-MB. (**F**) The expression and subcellular localization of EPSTI1 detected in muscle tissues from normal controls (NC1, NC2) and individuals with diabetes mellitus (DM1, DM2) . $^{**}P < 0.01$, $^{****}P < 0.0001$. EPSTI1: Epithelial-Stromal Interaction 1; DEGs: differentially expressed genes; IIMs: idiopathic inflammatory myopathies; CK-MB, Creatine kinase isoenzymes-MB

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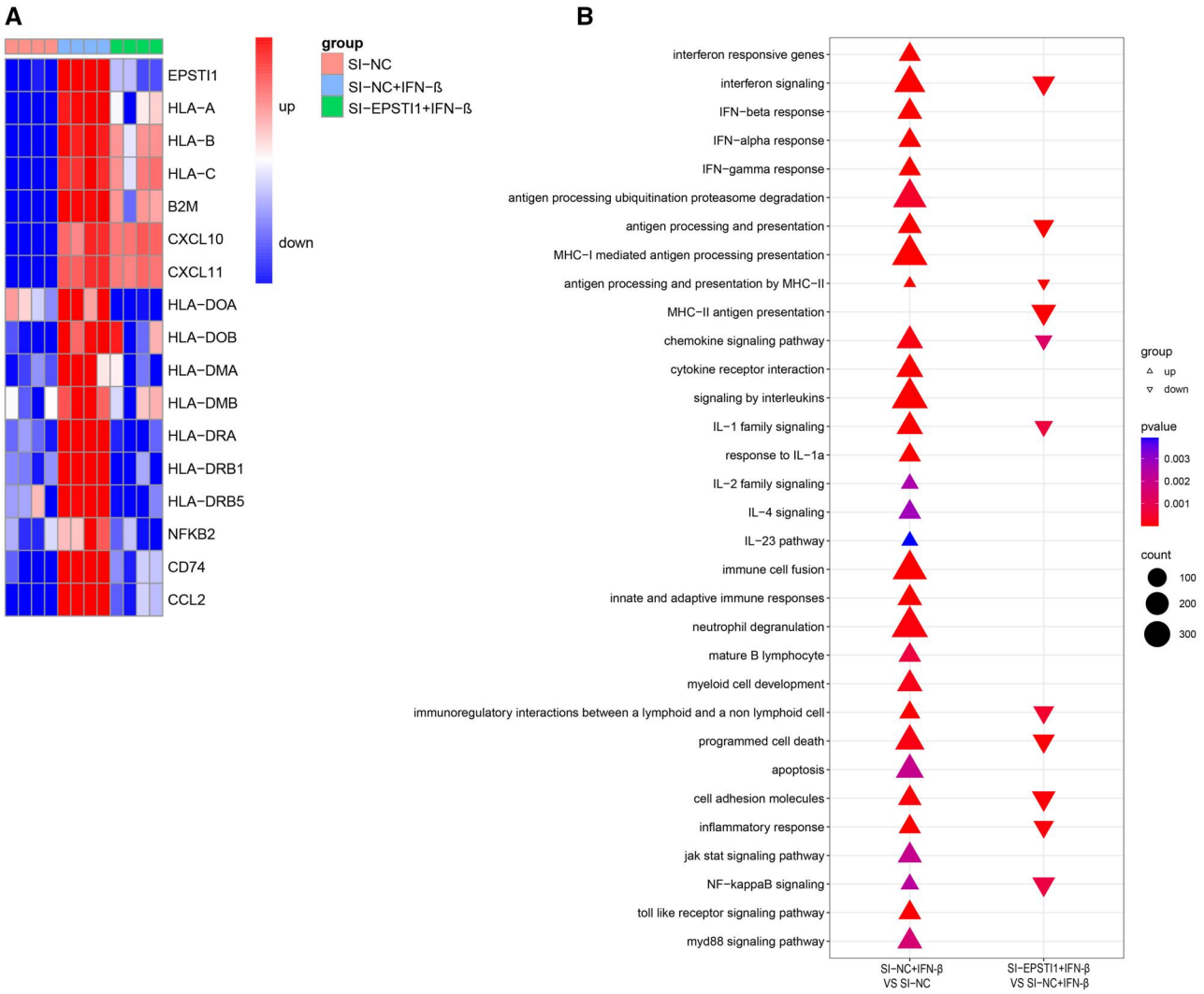

Figure 3. Transcriptome analysis of human myoblasts treated with si-EPSTI1 and IFN-β. Human myoblasts transfected with negative control siRNA or EPSTI1 siRNA for 24 h under the presence or absence of IFN-β (10 ng/ml). (**A**) Heatmap illustrates the expression of MHC-I class molecules (HLA-A, HLA-B, HLA-C, B2M), MHC-II class molecules (HLA-DOA, HLA-DOB, HLA-DMA, HLA-DMB, HLA-DRA, HLA-DRB), as well as chemokines CXCL10 and CXCL11. (**B**) The diagram illustrates the gene set enrichment analysis (GSEA) results comparing SI-NC and SI-NC+IFN-β, as well as SI-NC+IFN-β and SI-EPSTI1+IFN-β. EPSTI1: Epithelial-Stromal Interaction 1; siRNA: small interfering RNA

Similar to DM, studies have shown upregulation of EPSTI1 in peripheral blood and target tissues of patients with RA [28, 29], SLE [30, 31] and primary SS (pSS) [32, 33]. Both activation of the IFN pathway and upregulation of ISGs are common to these systemic autoimmune diseases. The IRGs, including EPSTI1, have been found to predict the response to rituximab treatment in RA patients [29]. The upregulation of EPSTI1 expression in PBMCs from SLE patients was not only inversely correlated with lymphocyte count, but was also identified to predict the efficacy of belimumab, suggesting at its potential as a biomarker for SLE onset and progression [30, 31]. EPSTI1 expression in pSS B cells was significantly higher than that of NC, which promoted Toll-like receptor 9 (TLR9) signalling activation and contributed to the abnormal B cell activation [32]. In our study, transcriptome analysis and *in vitro* studies of human myoblasts treated with si-EPSTI1 and IFN-β verified the involvement of EPSTI1 in IFN-β-induced HLA-A expression and chemokine-mediated signalling pathways in myoblasts.

EPSTI1 mRNA levels in muscle tissue positively correlated with CK-MB levels, which is typically associated with cardiac injury. However, studies have shown that elevated CK-MB levels in chronic inflammatory muscle diseases can be used to diagnose skeletal muscle damage instead of diagnosing myocardial damage in the absence of any evidence of myocardial ischaemia [34–38]. Meanwhile, EPSTI1 expression was predominantly observed in perifascicular atrophy. Since muscle weakness and pathological muscle damage did not correspond exactly [39], we hypothesized the involvement of EPSTI1 in muscle damage, in spite of the lack of correlation of it with MMT8. Therefore, the transcriptome analysis of human myoblasts treated with si-EPSTI1 and IFN-β was performed. Furthermore, a significant positive correlation was found between the expression of EPSTI1 in PBMCs and muscles. This expression pattern suggests that elevated EPSTI1 might be associated with systemic inflammation. As an ISG, EPSTI1 may correlate with disease prognosis like IFN [14], which requires further study.

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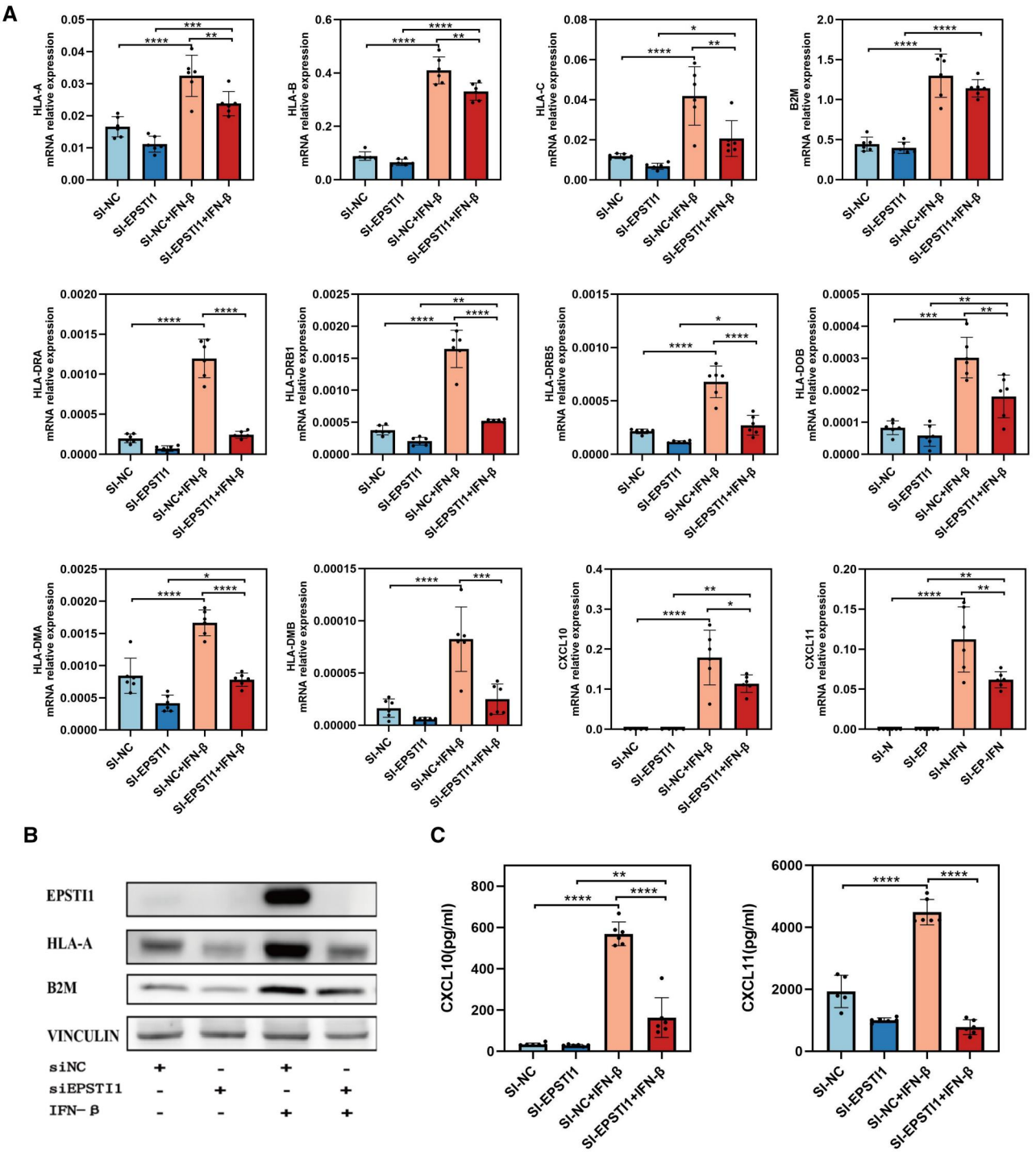

Figure 4. Inhibition of EPSTI1 prevented IFN-β induced antigen processing presentation and chemokine-mediated signalling pathways in myoblasts. (**A**) The mRNA expression levels of MHC class I (MHC-I) molecules (HLA-A, HLA-B, HLA-C, B2M) and MHC-II class molecules (HLA-DMA, HLA-DMB, HLA-DRA, HLA-DRB1, HLA-DRB5, HLA-DOB), as well as the chemokines CXCL10 and CXCL11 were quantified in human myoblasts transfected with negative control siRNA or EPSTI1 siRNA for 24 h in the presence or absence of IFN-β (10 ng/ml). (**B**) The protein expression levels of EPSTI1, HLA-A and B2M were assessed in each group of muscle cells. (**C**) The concentration of chemokines CXCL10 and CXCL11 secreted by myoblasts was measured using ELISA. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$. EPSTI1: Epithelial-Stromal Interaction 1; siRNA: small interfering RNA

In DM muscle tissue, a subset of myoblast progenitor cells show elevated levels of IFN-β and HLA-A, attracting immune responses. IFN-γ stimulation boosts HLA-A expression, which can be reduced by IFN-β neutralizing antibody, suggesting partial IFN-dependence of HLA-A upregulation [40]. Sciorati *et al.* found that MHC-I molecules appeared on muscle fibres a week after TLR agonist immunization, preceding immune cell infiltration, and were absent in IFN receptor–deficient mice [41]. In a TIF1-γ-induced myositis model, myofibril degeneration or atrophy coincided with MHC-I upregulation and immune cell infiltration [42]. Consistent with prior research, our study showed that IFN-β stimulation of myoblasts led to the upregulation of MHC-I class molecules and EPSTI1 expression. This highlights the regulatory

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role of type I IFN in the expression of MHC-I molecules, which may contribute to the pathogenesis of IIMs. Our sequencing results also revealed that EPSTI1 regulated several endoplasmic reticulum protein transport genes. These observations hint at a possible link between HLA-A expression and ER stress induced by EPSTI1, although further experimental confirmation is necessary.

Beyond MHC-I class molecules, the variations in CXCL10 and CXCL11 were particularly striking. CXCL10 also referred to as IFN-γ inducible protein-10 (IP-10), and CXCL11, known as IFN-induced T-cell alpha chemoattractant (I-TAC), have been associated with various autoimmune disorders and are secreted by several cell types, such as monocytes, endothelial cells and fibroblasts [43, 44]. Elevated CXCL10 and CXCL11 levels in DM and JDM patient serum correlate with disease activity [45, 46]. Anti-MDA5 DM patients have significantly higher CXCL10 levels than MDA5-negative patients, and these chemokines decrease after tofacitinib treatment [47, 48]. Additionally, CXCL10 and CXCL11 expression in C-protein-induced myositis (CIM) mice was observed within inflammatory infiltrates, while treatment of CIM mice with monoclonal anti-CXCL10 antibody significantly ameliorated muscle inflammation [49]. Given the above studies, we have not performed serum chemokine quantification in this study. Besides the pleiotropic cellular origins of circulating chemokines, our transcriptome sequencing and functional validation demonstrated it is regulatory effects on myocyte-derived chemokine production. This cell type–specific mechanistic insight would have been obscured by systemic serum measurements. In our study, IFN-β-stimulated myoblasts promote CXCL10 and CXCL11 secretion, and this effect could be attenuated by knocking down EPSTI1, indicating the fact that EPSTI1 regulates chemokines secretion.

In summary, our findings highlight the involvement of EPSTI1 in IFN-β-mediated HLA-A expression and chemokine secretion in myoblasts, potentially facilitating antigen presentation and inflammatory cell infiltration. This interaction between muscle cells and inflammatory cells may contribute to muscle damage. However, future studies are essential to elucidate further the mechanisms underlying the role of EPSTI1 in DM and to evaluate the potential efficacy of EPSTI1-targeted therapies in clinical trials.

Supplementary material

Supplementary material is available at *Rheumatology* online.

Data availability

Data are available upon reasonable request.

Contribution statement

H.L. and H.Z. designed the research study. X.L., T.D., Y.X., J.Z., T.H., S.X., Q.M., and W.H. performed the research. X. L. and T.D. analyzed the data and wrote the manuscript. All authors read and approved the final manuscript.

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